

Assignment 3: Human Activity Recognition

April 23, 2026

Data Mining
Spring 2026

Assignment 3

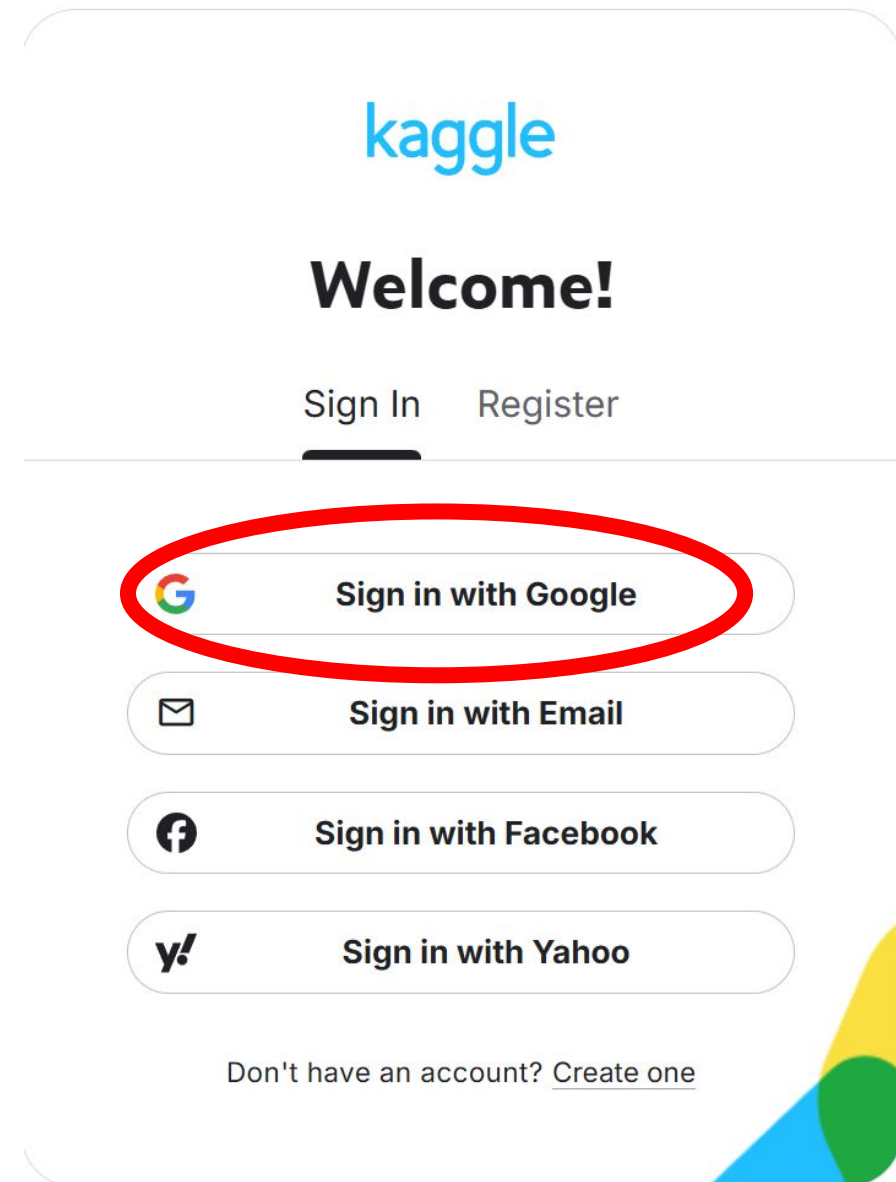
The Assignment 3 will be conducted as a Kaggle Competition.

- It is an **individual work** and counts for **20%** of the semester grade
- [Assignment 3 Kaggle Link](#)
- **Deadline:**
 - **June 10, 11:55 pm**
- **Task:**
 - **Accelerometer Readings for Human Activity Recognition (HAR)**
 - Use data from individuals wearing accelerometers to predict each user's activity in 5-minute intervals.

Kaggle Basic Setup

- **Account Setup**

Please sign in with your **NYCU email** account directly.



Kaggle Basic Setup

- **(IMPORTANT) Display Name Change**
before Kaggle submission

Path: Go to the competition page → **Team**
→ **Your Team** → **Team Name**

Format: StudentID (e.g., 314551130)

Note: Teams that do not use a valid Student ID as their name or sign in without NYCU email account will be **removed** from the leaderboard without prior notice!!

NYCU Data Mining Assignment 3 (2026 Spring)

NYCU Data Mining Assignment 3

Overview Data Discussion Leaderboard Rules **Team** Submissions

Your Team

Everyone that competes in a Competition does so as a team - even if you're competing by yourself

General

TEAM NAME

314551130

This name will appear on your team's leaderboard position.

Kaggle Basic Setup

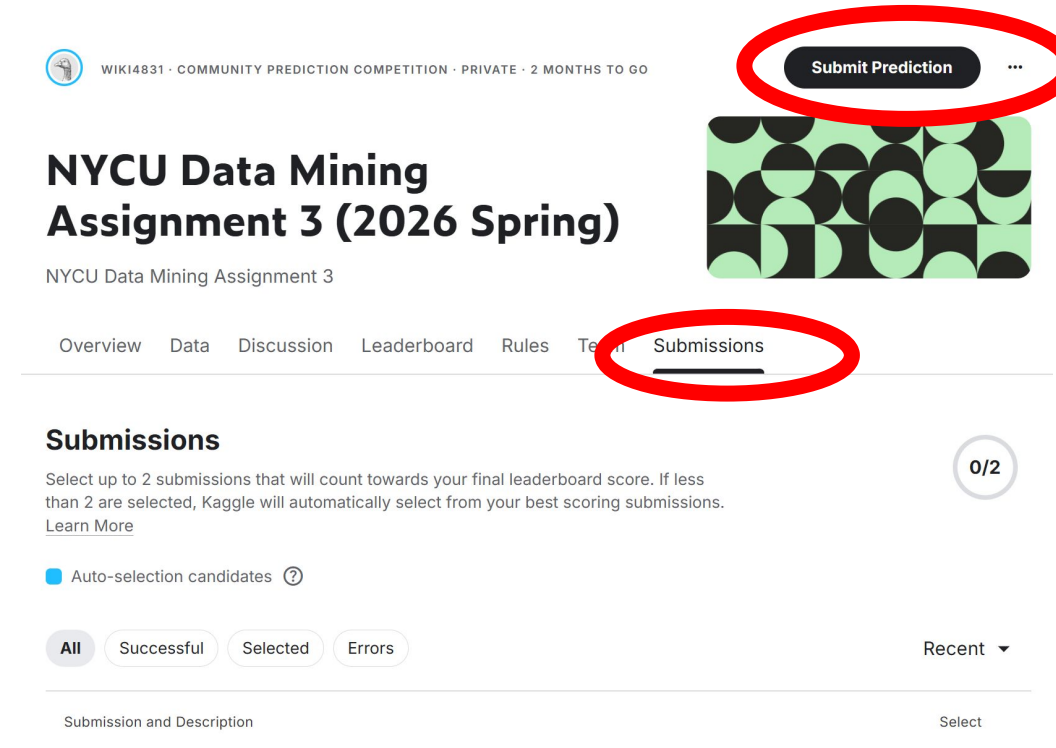
- **How to Submit**

Upload:

Click the **Submit Predictions** button at the top right of the competition page.

Checking Results:

You can view your history and the status of your uploads under the Submissions tab.



Kaggle Basic Setup

- **Leaderboard**

Public Leaderboard: Your score on a subset of the test data is shown **immediately** after submission.

Private Leaderboard: The results for the remaining hidden data will only be revealed **after the competition ends**.

Dataset Overview

Accelerometer Readings from 100 users

The original raw data was recorded at a frequency higher than 1 Hz. The provided data has been aggregated into 1-second intervals. Each row represents statistics for one second.

- **Columns:**
 - **index:** The sequential order of each second within the 5-minute sequence (0–299).
 - **mean_x, mean_y, mean_z:** Average acceleration along the X, Y, and Z axes in a 1-second interval.
 - **std_x, std_y, std_z:** Standard deviation of acceleration along the X, Y, and Z axes in a 1-second interval.
 - **file_id:** The unique global file ID. (Note: Use this ID in your sample_submission.csv).
- **Label:** Ranges from 0 to 5 (indicating different **Human Activity**).
 - Note: Each 5-minute time window (each CSV file) is assigned a single activity label.

Dataset Overview

Folder Organization

- **train/**& **test/**: Contains training data grouped by individual users (**User_xxx**).

Data Content

Each file contains sequential statistical features (Mean/Std),
For each single CSV file(with same file id) corresponds to **exactly one**
specific activity label.



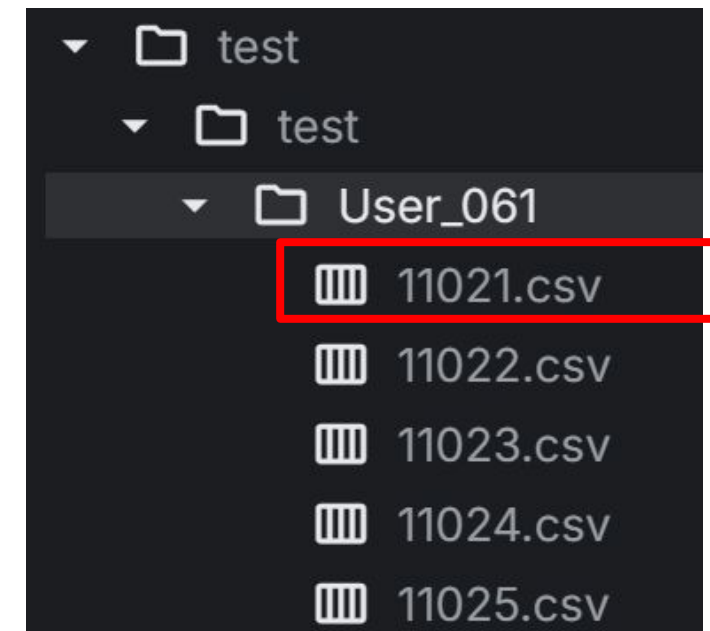
index	mean_x	mean_y	mean_z	std_x	std_y	std_z	label	file_id
0	-0.4676289664	-0.5378539625	0.6572404470000001	0.0037338272032503	0.0070966772756883	0.0047337407424678	0	1
1	-0.4666903600000001	-0.5470351	0.6549311731999999	1.1158161231197418e-16	0.0050822829679846	0.0065114227135875	0	1
2	-0.4673160976000001	-0.5359866124999999	0.6584720598000001	0.0030809189658331	0.0058747635216081	0.0021881713389351	0	1
3	-0.4688804416000001	-0.5333412	0.65847206	0.005455416425332	0.0	0.0	0	1
4	-0.4702883512000001	-0.5333412	0.65847206	0.0066164333813123	0.0	0.0	0	1
5	-0.4674725320000001	-0.5422111125	0.6572404472	0.0034265846756633	0.0077428088587541	0.0041976431608682	0	1

Kaggle Submission Format

Format: We provide a template file in the Data tab (`sample_submission.csv`).

- Includes two columns: **Id** and **Label**
- Each row represents the prediction result of a CSV file (5-minute time window) under the “test” folder

1	Id	,	Label
2	11021,		0
3	11022,		0
4	11023,		0
5	11024,		0
6	11025,		0



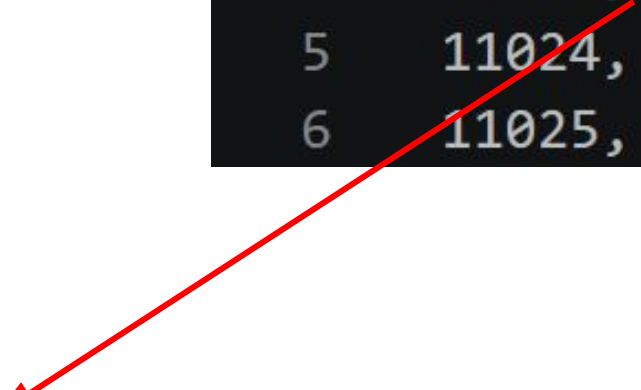
Kaggle Submission Format

Format: We provide a template file in the Data tab (`sample_submission.csv`).

- Includes two columns: **Id** and **Label**

Requirements:

- Please fill your predictions in the **Label** field (label 0 to 5).



	Id	,	Label
1	11021,		0
2	11022,		0
3	11023,		0
4	11024,		0
5	11025,		0
6			0

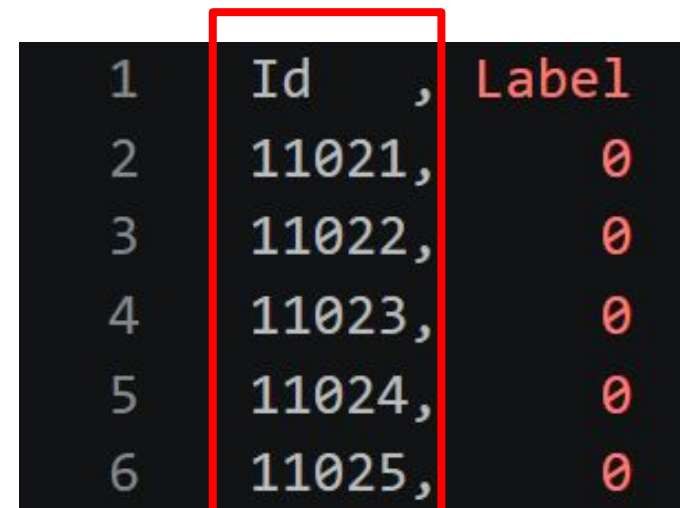
Kaggle Submission Format

Format: We provide a template file in the Data tab (`sample_submission.csv`).

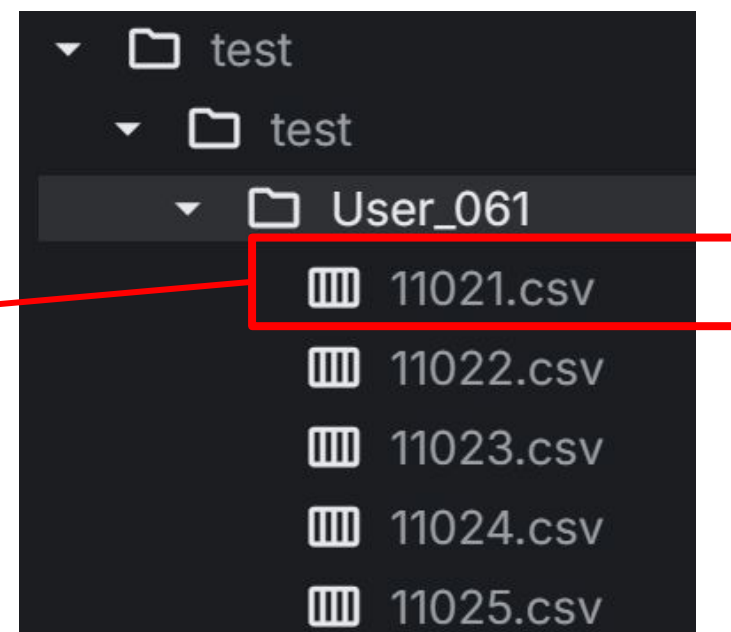
- Includes two columns: **Id** and **Label**

Requirements:

- Each **Id** corresponds to exactly one CSV file in the test set.
 - E.g, 11021
 - Also “file_id” in each CSV file



1	Id	,	Label
2	11021,		0
3	11022,		0
4	11023,		0
5	11024,		0
6	11025,		0



Kaggle Submission Format

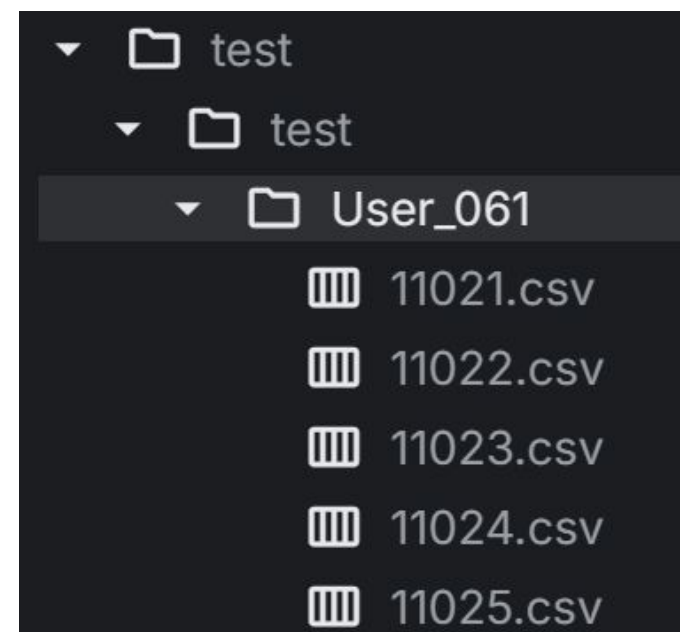
Format: We provide a template file in the Data tab (`sample_submission.csv`).

- Includes two columns: **Id** and **Label**

Requirements:

- **Crucial:** Ensure your submission formats are strictly aligned with the IDs provided; any mismatch will result in an error or poor performance.

1	Id	,	Label
2	11021,		0
3	11022,		0
4	11023,		0
5	11024,		0
6	11025,		0



Kaggle Submission Rule

- **Evaluation:** Submissions are evaluated using **F1-score (macro)**.
- **Daily Limit:** You are allowed **3 submissions per day**. Use them wisely!
- The private leaderboard will only be visible after the competition ends.
 - By default, your **two** best public leaderboard submissions will be used for the private leaderboard.

Grading Criteria - Kaggle Competition (60%)

The grade for Assignment 3 is based on the **Kaggle competition** and **your report**.

- **Kaggle Competition (60%)**
 - 30% based on Public Leaderboard Points (50% of testing data)
 - 70% based on Private Leaderboard Points (remaining 50% of testing data)
- **Three Baselines**

Public Leaderboard

(Private Leaderboard is hidden until the end of the competition)

#	Team	Members	Score
	Baseline-3		0.7088
	Baseline-2		0.6130
	Baseline-1		0.1201

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Leaderboard Point Calculation

Your Model Performance	Points	Calculation Method	N: the number of people over Baseline 3 (If N=1, then the point = 60)
Above Baseline 3	42 ~ 60	$42 + 18 \times \frac{N - \text{Rank}}{N - 1}$	
Between Baselines 2 & 3	25 ~ 40	$25 + 15 \times \frac{\text{YOUR_Score} - \text{B2_Score}}{\text{B3_Score} - \text{B2_Score}}$	
Between Baselines 1 & 2	10 ~ 25	$10 + 15 \times \frac{\text{YOUR_Score} - \text{B1_Score}}{\text{B2_Score} - \text{B1_Score}}$	
Under Baseline 1	5	5	

Grading Criteria - Kaggle Competition (60%)

○ Example

■ Public Learboard Points

- Assume your public score is 0.65 (between Baseline 2 & Baseline 3)
- Your Public Points:

- $25 + 15 * (0.65 - 0.6130) / (0.7088 - 0.6130) = 30.7933$

$$25 + 15 \times \frac{\text{YOUR_Score} - \text{B2_Score}}{\text{B3_Score} - \text{B2_Score}}$$

■ Private Learboard Points

- Assume your private score is above Baseline 3, and your rank is 2nd (Rank = 2).
- There are 11 students above Baseline 3 (N = 11).
- Your Private Points:

- $42 + 18 * (11 - 2) / (11 - 1) = 58.2$

$$42 + 18 \times \frac{N - \text{Rank}}{N - 1}$$

■ Your Kaggle Competition Score:

- $0.7 * 58.2 + 0.3 * 30.7933 = 49.978$

■ Your Final Assignment 3 score = 49.978 + Report Score

Public Leaderboard

#	Team	Members	Score
	Baseline-3		0.7088
	Baseline-2		0.6130
	Baseline-1		0.1201

Grading Criteria - Kaggle Competition (60%)

The grade for Assignment 3 is based on the **Kaggle competition** and **your report**.

- **The private test data is typically more challenging than the public test data.**
 - To avoid overfitting to the public leaderboard, it is recommended to create validation sets from the training data for model development.

Grading Criteria - Report (40%)

The grade for Assignment 3 is based on the **Kaggle competition** and **your report**.

- **Report (40%)**

- **Basic Requirements**

(If any of the following information is missing, **the grade for Assignment 3 will be zero.**)

- Please provide the public GitHub link to your Assignment 3 code.
 - Please describe in detail how to run your code.

Grading Criteria - Report (40%)

The grade for Assignment 3 is based on the **Kaggle competition** and **your report**.

- **Report (40%)** **More detailed descriptions and analysis will receive higher scores**
 - **Please answer the following questions**
 1. **(10%)** Please provide a preliminary analysis that informs your method design (e.g., observations from the raw data, performance of naive baseline methods, etc.).
 2. **(10%)** What preprocessing techniques did you use to improve performance? How much improvement did each technique achieve?
 3. **(10%)** Please describe how you align the activity labels with the sequential accelerometer readings (e.g., what temporal features you added or how your model captures temporal dependencies).
 4. **(10%)** Please provide an ablation study for your core technical design choices.

Grading Criteria

- You are allowed to use the AI tools to assist your work, but ...
 - Any form of plagiarism in code or report will result in a **score of 0 for this assignment.**
 - The submitted **report, code, and Kaggle results must be consistent and reproducible.**
 - If the code cannot be executed, or if the Kaggle results are inconsistent with the submitted code or report, this will also result in a **score of 0 for this assignment.**

**If there are concerns about your submission,
TAs may require an on-site demonstration to verify the results.**

Report and Code Submission

Please submit your report & code by **June 10, 11:55 PM.**

Late submissions are not accepted.

Submissions consist of two parts (Besides Kaggle Submission):

1. Code

- a. Please host your **code on a public GitHub repository** (commit before deadline).

2. Report.

- a. Please ensure your public **GitHub link is included** within your report, and submit your **report via E3.**
- b. **File name:** DM_asg3_{studentID}.pdf (e.g., DM_asg3_314551051.pdf)

Please write down your report in **English.**

The TA will test whether your code can run successfully; additionally, your **Kaggle Display Name** must be your **Student ID**, or your grade will be recorded as 0.

If you have any other questions, please feel free to contact the TA at nycu.dm.ta@gmail.com.

Additional Information

Following provide more detailed information for the 3-axis accelerometer in the provided dataset:

- The unit of the measurements is g (gravitational acceleration), specifically, $1g \approx 9.81 \text{ m/s}^2$
 - The data does not remove the gravity components.
- The 3-axis accelerometer is worn on users' wrists
 - The Fig 3. in [1] provides an example

Note that

- The current three baselines on Kaggle did not use the above information for training and testing.
- The above information is for reference only. Including it is optional.

[1] Bruno, Barbara, Fulvio Mastrogiovanni, and Antonio Sgorbissa. "A public domain dataset for ADL recognition using wrist-placed accelerometers." *the 23rd IEEE International Symposium on Robot and Human Interactive Communication*. IEEE, 2014.